Bayes and Kalman Filter

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Abstract - State estimation enables autonomous systems to infer exact physical states from imperfect sensor measurements. This paper provides a comprehensive breakdown of the Recursive Bayes Filter and the Linear Kalman Filter (KF). We unpack core probabilistic terms using physical intuition and analyze the three fundamental assumptions (Linearity, Gaussian Noise, and Gaussian Initial Belief) that make the Bayes Filter computationally tractable. Additionally, we detail the five standard Kalman Filter equations and their role in balancing model predictions with noisy observations. Finally, we showcase its operational integration within a Formula Student Driverless (FSD) race car pipeline to overcome severe disturbances.

I. Theoretical Foundations of Bayes Filter

Imagine driving a race car through a dense fog while blindfolded, relying solely on two imperfect sources of information: your internal feel for how fast you pressed the gas pedal 5 seconds ago, and occasional, noisy shouts from a passenger reading GPS coordinates. How do you combine your physical intuition of how cars move with noisy sensor readings to pinpoint your exact location on a race track?

This fundamental challenge is known as State Estimation. In physical autonomous systems, such as self driving cars, drone quadcopters, or mobile warehouse robots, the true internal state vector xt (encompassing 2D position, heading orientation, forward velocity, slip angle, and rotational yaw rate) can never be measured directly with absolute perfection. Instead, onboard hardware sensors—such as wheel speed encoders, Inertial Measurement Units (IMUs), LiDAR scanners, stereo cameras, and GPS receivers—provide corrupted, partial, and delayed observations tangled with physical environmental noise.

Probabilistic state estimation techniques provide the mathematical framework to convert these unreliable streams of raw data into optimal, highly accurate estimates of system behavior [1]. Rather than guessing a single fixed point estimate, probabilistic filters continuously track and update a complete probability distribution over all possible vehicle positions over time.

II. Theoretical Foundations of the Recursive Bayes FIlter

The recursive Bayes filter represents the overarching theoretical framework for all dynamic state estimation algorithms. Its primary objective is to calculate the posterior probability density function of the state xt at discrete time step t, conditioned on the complete historical sequence of all observations y1:t and control inputs u1:t:

$$p(\mathbf{x}_t \mid \mathbf{y}_{1:t}, \mathbf{u}_{1:t})$$

A. Core Independence Assumptions

To compute this conditional probability in real time on embedded microprocessors, the recursive Bayes filter relies on two fundamental structural assumptions, modeled as a Hidden Markov Model (HMM):

1) First-Order Markov Property: The current physical state xt depends strictly on the immediately preceding state xt-1 and the current control input ut. If we know exactly where the car was a fraction of a second ago and how hard we turned the steering wheel, historical states prior to step t-1 provide no additional information about the present state:

$$p(\mathbf{x}_t \mid \mathbf{x}_{1:t-1}, \mathbf{y}_{1:t-1}, \mathbf{u}_{1:t}) = p(\mathbf{x}_t \mid \mathbf{x}_{t-1}, \mathbf{u}_t)$$

Conditional Measurement Independence: The current sensor reading yt depends exclusively on the true current state xt. Given the true state of the vehicle right now, past measurements or past control signals provide zero direct information about the current sensor reading:

$$p(\mathbf{y}_t \mid \mathbf{x}_{1:t}, \mathbf{y}_{1:t-1}, \mathbf{u}_{1:t}) = p(\mathbf{y}_t \mid \mathbf{x}_t)$$

B. Two-Stage Recursive Architecture

The Bayes filter executes continuously as a cyclic, two step recursion: state prediction followed by measurement update.

Step 1: Predict (State propagation)

In the prediction phase, the filter projects the probability distribution forward in time using the physical equations of motion, before looking at any new sensor data. Using the law of total probability and marginalizing over all possible prior states x(t-1).

$$p(\mathbf{x}_t \mid \mathbf{y}_{1:t-1}, \mathbf{u}_{1:t}) = \int p(\mathbf{x}_t \mid \mathbf{x}_{t-1}, \mathbf{u}_t)\, p(\mathbf{x}_{t-1} \mid \mathbf{y}_{1:t-1}, \mathbf{u}_{1:t-1})\, d\mathbf{x}_{t-1}$$

Let us break down each physical term inside this prediction integral:

$$p(\mathbf{x}_{t-1} \mid \mathbf{y}_{1:t-1}, \mathbf{u}_{1:t-1})$$

Prior belief: Our absolute best estimate of the vehicle's position at the previous time step, after accounting for all past measurements and actions.

$$p(\mathbf{x}_t \mid \mathbf{x}_{t-1}, \mathbf{u}_t)$$

Motion Model: The probabilistic kinematic model of the vehicle. It describes: "If the car was at position x_{t-1} and we applied steering/accelerator input u_t, where do physics dictate the car will land at time t?"

$$p(\mathbf{x}_{t-1} \mid \mathbf{y}_{1:t-1}, \mathbf{u}_{1:t-1})$$

Predicted Belief (Prior for Update): The overall forecast of the state at time t based purely on motor commands and physical kinematics, prior to checking our sensors. Because physical movement introduces friction, uneven road surfaces, and wheel slip, this prediction step always increases our state uncertainty (widens the probability curve).

Step 2: Update (Measurement Correction)

When a new sensor measurement y arrives (e.g., a camera visual odometry check or GPS ping), the filter uses Bayes' Rule to refine the predicted belief:

$$p(\mathbf{x}_t \mid \mathbf{y}_{1:t}, \mathbf{u}_{1:t}) = [\, p(\mathbf{y}_t \mid \mathbf{x}_t)\, p(\mathbf{x}_t \mid \mathbf{y}_{1:t-1}, \mathbf{u}_{1:t}) \,] / \eta$$

where the normalization scaling factor η is given by:

$$\eta = \int p(\mathbf{y}_t \mid \mathbf{x}_t)\, p(\mathbf{x}_t \mid \mathbf{y}_{1:t-1}, \mathbf{u}_{1:t})\, d\mathbf{x}_t$$

Unpacking the physical update terms:

- *Measurement Likelihood $p(y_t|x_t)$:* The generative sensor model. It answers: "If the car were truly located at state x_t, how likely is it that our camera or LiDAR would output the exact measurement y_t?"
- *Normalizer (Marginal Likelihood) η:* A mathematical scaling factor that integrates across all candidate physical states to ensure that the total area under the posterior probability curve equals exactly 1.0 (100% total probability).
- *Posterior Belief $p(x_t \mid y_{1:t}, u_{1:t})$:* The final refined state probability density. By combining physical predictions with sensor observations, the measurement update reduces uncertainty (sharpens the probability curve)

III. Reduction To The Linear Kalman Filter

While the general Recursive Bayes Filter (Section II) provides a complete mathematical framework, evaluating its continuous integrals across multi-dimensional continuous spaces in real time is computationally impossible on digital microprocessors. Performing continuous integration across arbitrary probability distributions would require infinite computing power.

The Linear Kalman Filter (LKF) is an exact analytical reduction of the Bayes Filter under three specific structural assumptions that turn complex continuous integrals into lightning-fast matrix arithmetic.

A. The Three Fundamental Reduction Assumptions

1) Discrete-Time Linear System Dynamics: The vehicle physics (kinematics) and the sensor observation dynamics must be purely linear matrix functions:

$$\mathbf{x}_t = \mathbf{A}_{t-1}\, \mathbf{x}_{t-1} + \mathbf{B}_t\, \mathbf{u}_t + \mathbf{w}_{t-1} \quad (6)$$

$$\mathbf{y}_t = \mathbf{C}_t\, \mathbf{x}_t + \mathbf{D}_t\, \mathbf{u}_t + \mathbf{v}_t \quad (7)$$

where A_{t-1} is the state transition matrix, B_t is the control input matrix, C_t is the measurement mapping matrix, and D_t is the direct feedthrough matrix.

2) Additive Zero-Mean Gaussian White Noise: Unmodeled physical disturbances (process noise wt, such as wind gusts or engine vibrations) and physical sensor corruptions (measurement noise v_t) are modeled as uncorrelated, zero-mean multivariate Gaussian distributions:

$$\mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_t), \quad \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_t) \quad (8)$$

where Q_t is the positive semi-definite process noise covariance matrix and R_t is the positive definite measurement noise covariance matrix.

3) Gaussian Initial State Belief: The starting estimate of the vehicle state prior to turn-on, p(x0), is modeled as a Gaussian distribution parameterized entirely by an initial mean state vector x^0 and an initial error covariance matrix P_0:

$$p(\mathbf{x}_0) = \mathcal{N}(\hat{\mathbf{x}}_0, \mathbf{P}_0) \quad (9)$$

B. Mathematical Tractability: Why Gaussians make the math easy

Why do these three assumptions make state estimation computationally tractable? The answer lies in two unique linear-algebraic properties of Gaussian distributions:

1. *Linear Closure under Transformation:* When a Gaussian distribution passes through a linear transformation (matrix multiplication and addition), the output distribution remains 100% strictly Gaussian.
2. *Product of Gaussians*: Multiplying two Gaussian distributions together (Bayes' rule update step: multiplying predicted state by sensor likelihood) produces another exact single Gaussian distribution.
3. *Finite Parameterization*: An arbitrary continuous probability curve requires an infinite list of points to describe its shape. However, a Gaussian distribution in N dimensions is completely described by just two

finite matrices: a Mean Vector $\hat{x}$ (center of mass) and a Covariance Matrix P (spread/uncertainty).

Because of these properties, continuous integrals (3) and (5) disappear entirely! The infinite-dimensional Bayes filter simplifies into deterministic matrix additions, multiplications, and inversions executing in fractions of a millisecond.

IV. Detailed Specification And Equation Breakdown

The linear Kalman filter operates recursively across time index step n using five core matrix equations. We adopt standard operational notation: $\hat{x}_{n|n-1}$ represents the prior predicted state estimate for step n given data up to step n 1, while $\hat{x}_{n|n}$ represents the posterior updated state estimate after fusing the sensor measurement at step n.

A. Summary Breakdown of the Predict-Update Lifecycle

The complete recursive pipeline in three (sometimes considered 2) continuous phases.

1) Initialization Phase (Step n = 0):

- Set initial state mean: $\hat{x}_{0|0} = z_0$
- Set initial error covariance: $P_{0|0} = R_0$

2) Prediction Phase (Time Update/ State Propagation):

- State Extrapolation:
$$\hat{x}_{n|n-1} = F\,\hat{x}_{n-1|n-1} + G\,u_{n-1}$$
- Covariance Extrapolation:
$$P_{n|n-1} = F\,P_{n-1|n-1}\,FT + Q$$

3) Update Phase (Measurement Correction & Gain Calculation):

- Kalman Gain Computation:
$$K_n = P_{n|n-1}\,H^T\,(H\,P_{n|n-1}\,H^T + R_n)^{-1}$$
- State Estimate Correction:
$$\hat{x}_{n|n} = \hat{x}_{n|n-1} + K_n\,(z_n\,H\,\hat{x}_{n|n-1})$$
- Covariance Correction (Joseph Form):
$$P_{n|n} = (I\,K_n\,H)\,P_{n|n-1}\,(I - K_n\,H)^T + K_nR_nK_n^T$$

B. Detailed Mathematical & physical Analaysis of the 5 Equations

Equation 1: State Extrapolation (Predict State Mean)

$$\hat{x}_{n|n-1} = F\,\hat{x}_{n-1|n-1} + G\,u_{n-1} \quad (10)$$

Physical Meaning: This equation projects our vehicle's state forward in time using basic high-school physics. Matrix F (State Transition Matrix) projects previous position and speed forward using kinematic relationships (e.g., $x_{new} = x_{old} + v \cdot \Delta t$). Matrix G (Control Matrix) converts physical driver commands u_{n-1} (such as pushing the throttle pedal or turning the steering wheel) into adjustments in acceleration and velocity.

Equation 2: Covariance Extrapolation (Predict State Uncertainty)

$$P_{n|n-1} = F\,P_{n-1}|_{n-1}\,F^T + Q \quad (11)$$

Physical Meaning: This equation calculates how our estimate uncertainty grows while moving blind. The term $F\,P_{n-1|n-1}\,F^T$ passes prior uncertainty through the kinematic equations. Because our physical motion model is never 100% perfect, we add matrix Q (Process Noise Covariance). Matrix Q injects uncertainty to account for unmodeled real-world phenomena like wind drag, tire slippage, and road bumps.

Equation 3: Kalman Gain Computation

$$K_n = P_{n|n-1}\,H^T\,(H\,P_{n|n-1}\,H^T + R_n)^{-1} \quad (12)$$

Physical Meaning: The Kalman Gain matrix Kn is the absolute core of the filter. It acts as an adaptive trust weighting factor between 0.0 and 1.0. Matrix H translates state space variables into physical sensor dimensions. Matrix R_n represents physical sensor noise variance. The denominator ($H\,P_{n|n-1}\,H^T + R_n$) represents total system innovation uncertainty—combining model prediction error with physical sensor noise.

Equation 4: State Update (Measurement Correction)

$$\hat{x}_{n|n} = \hat{x}_{n|n-1} + K_n\,(z_n - H\,\hat{x}_{n|n-1}) \quad (13)$$

Physical Meaning: This equation updates our vehicle position estimate by fusing physical prediction with incoming sensor readings z_n. The term ($z_n - H\,\hat{x}_{n|n-1}$) is called the Measurement Innovation (or Residual). It measures the discrepancy between what our physical sensor actually sees vs. what our kinematic model predicted. The Kalman Gain K_n scales this residual error to apply a optimal correction back to the state vector.

Equation 5: Covariance Update (Correction)

$$P_{n|n} = (I - K_n\,H)\,P_{n|n-1}\,(I - K_n\,H)^T + K_n\,R_n\,K_n^T \quad (14)$$

Physical Meaning: Incorporating new sensor data provides new information, which shrinks our state uncertainty. While a simplified equation exists ($P_{n|n} = (I - K_n\,H)\,P_{n|n-1}$), real digital hardware suffers from floating-point rounding errors. Equation (14)—known as the Joseph Form Formulation—guarantees that matrix $P_{n|n}$ remains mathematically symmetric and positive-definite, preventing software crashes in production autonomous vehicles

C. Intuitive Behavior of Kalman Gain in Extreme Sensor Regimes

To understand how the Kalman Gain dynamically balances trust between physics predictions and sensor readings, let us examine scalar simplification where observation matrix H = 1:

$$K_n = P_{n|n-1} / (P_{n|n-1} + R_n) \quad (15)$$

Extreme Case 1: Extremely Noisy Sensor ($Rn \to \infty$)

Imagine driving through heavy rain where your visual sensors produce corrupted, wild measurements. Sensor noise variance Rn becomes extremely large. Taking the limit as $R_n \to \infty$:

$$\lim_{R_n \to \infty} K_n = \lim_{R_n \to \infty} [\, P_{n|n-1} / (P_{n|n-1} + R_n) \,] = 0 \quad (16)$$

Substituting Kn = 0 back into State Update Equation (13):

$$\hat{\mathbf{x}}_{n|n} = \hat{\mathbf{x}}_{n|n-1} + 0 \cdot (\mathbf{z}_n - \hat{\mathbf{x}}_{n|n-1}) = \hat{\mathbf{x}}_{n|n-1}$$

Result: The filter completely ignores incoming noisy sensor data z_n and relies 100% on dynamic physical physics prediction $\hat{x}_{n|n-1}$.

Extreme Case 2: Ultra-Accurate Sensor ($R_n \to 0$)

Now imagine operating with a high-precision sensor with zero noise variance ($R_n \to 0$). Taking the limit as $R_n \to 0$:

$$\lim_{R_n \to 0} K_n = P_{n|n-1} / (P_{n|n-1} + 0) = 1 \quad (17)$$

Substituting $K_n = 1$ back into State Update Equation (13):

$$\hat{\mathbf{x}}_{n|n} = \hat{\mathbf{x}}_{n|n-1} + 1 \cdot (\mathbf{z}_n - \hat{\mathbf{x}}_{n|n-1}) = \mathbf{z}_n$$

Result: The filter completely bypasses kinematic physics predictions and forces the estimated state to equal the raw sensor measurement Z_n.

V. Formula Student Driverless Application

A. Formulating the 2D Racing State Vector

For an FSD race car operating on a 2D track surface, the state vector $x_t \in \mathbb{R}6$ is defined as:

$$x_t = [x, y, \theta, v_x, v_y, \omega]^T \quad (18)$$

where *(x, y)* represents global Cartesian track coordinates, θ is vehicle yaw heading orientation, v_x is longitudinal velocity along the car chassis axis, v_y is lateral slip velocity, and $\omega = \dot{\theta}$ is rotational yaw rate

B. Multi-Sensor Data Fusion Pipleline Architecture

The Kalman Filter acts as a centralized data fusion hub, combining sensor streams operating at vastly different sampling frequencies:

1) High-Rate Kinematic Prediction (Proprioceptive Sensing):

a) Optical Wheel Encoders

Measure individual wheel rotation speeds to calculate longitudinal vehicle reference speed vwheel at 100 Hz.

b) Inertial Measurement Unit (IMU)

Measures chassis longitudinal/lateral accelerations (a_x, a_y) and yaw rate ω at high sampling rates (200 Hz 500 Hz).

c) State Propagation:

At every microsecond IMU tick, the filter executes Prediction Equations (10) and (11) to dead-reckon vehicle pose forward in real time.

2) Low-Rate Corrective Measurement Updates (Exteroceptive Sensing):

a) LiDAR Scanner & Stereo Cameras:

Run Simultaneous Localization and Mapping (SLAM) algorithms to detect track border cones. Visual odometry generates global pose corrections $z_{pose} = [x_{slam}, y_{slam}, \theta_{slam}]^T$ at lower rates (10 Hz - 30 Hz).

b) GNSS / GPS Module:

Provides global position updates $z_{gps} = [x_{gps}, y_{gps}]T$ in open outdoor tracks.

c) Measurement Fusion

Whenever a low-rate perception packet arrives, observation matrix H maps state space variables to sensor outputs, executing Update Equations (12)-(14) to eliminate dead-reckoning position drift

C. Resolving Real-World Autonomous Racing Challenges

1) Tire Slip & High Lateral G-Forces: High-speed cornering causes race car tires to slip laterally, violating rigid rolling assumptions. By dynamically expanding lateral process noise covariance Q_{v_y}, the filter prevents severe estimation bias, allowing the vehicle to accurately track side-slip drift angle during aggressive cornering.

2) Sensor Jitter and Dust Outliers: Track dust, tire debris, and vehicle chassis vibration introduce extreme sensor outliers into LiDAR and camera feeds. The Kalman filter evaluates the Mahalanobis Innovation Distance:

$$d^2 = (\mathbf{z}_n - \mathbf{H}\,\hat{\mathbf{x}}_{n|n-1})^T \mathbf{S}_n^{-1} (\mathbf{z}_n - \mathbf{H}\,\hat{\mathbf{x}}_{n|n-1})$$

If a sensor reading exceeds a critical threshold, it is flagged as an outlier. The filter temporarily forces $K_n \to 0$, discarding the corrupted measurement and safely relying on high-speed IMU dead-reckoning until perception recovers.

3) Downstream Control Enablers: The output updated state estimate $\hat{x}_{n|n}$ provides clean, low-latency velocity and position estimates to downstream Model Predictive Controllers (MPC) and Torque Vectoring algorithms, enabling precise throttle, steering, and braking control at the handling limits of the race car.

VI. Conclusion

This report has presented a unified, accessible breakdown of probabilistic state estimation, spanning from foundational Recursive Bayes Filter theory to real-world embedded deployment in autonomous driverless vehicles. By establishing the linear-Gaussian reduction assumptions, we demonstrated how continuous integrals reduce to robust linear algebra operations. The detailed analysis of the five Kalman Filter equations and extreme gain behaviors provides early-stage engineering students with a solid foundation for designing autonomous robotics pipelines.

VII. Acknolwedgement

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VIII. References

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